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课题研究报告 · RESEARCH STUDY

AI 产品工程：从演示到可验证结果的产品学科

AI Product Engineering: The Product Discipline from the Demo to the Verified Outcome

关于评估规格、生产仪表、吸收线与智能边际成本的课题研究

A research study on evaluation as specification, production instruments, the absorption line and the marginal cost of intelligence

研究底本 SOURCE
Cuofano, G. AI Product Engineering — The Product Discipline of the AI Era (2026)

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摘要 *Abstract*

许多 AI 产品在演示中出色，却在生产中失败。本研究以 Cuofano（2026）《AI Product Engineering》为底本，追问：当产品的核心组件变为概率性的、租用的、且频繁更换的模型时，产品学科应如何重建？研究采用结构化文本分析，将底本三十一章、八个公开案例与一个复合案例拆解为机制、工具与案例三类分析单位，并据底本自列的证伪条件形成六项假设。主要发现有四：其一，**评估即规格**——由业务数字、领域专家签署的黄金集与评估管线构成的套件取代了需求文档，模型因而可以“最后选择”；其二，可观测性、数据底座、编排与治理构成生产仪表，多数生产失败可归入可观测性、评估与治理三类缺口；其三，**吸收线**把路线图分为易逝的补偿与持久的要求，持久价值位于企业自有的标准、案例、数据与决策节点，并经由人类与代理双界面交付；其四，智能为软件带回了**边际成本**，毛利率成为设计结果，定价由席位转向用量与结果。八个案例在不同层级上印证了这些机制。文末讨论了研究局限并提出后续研究议程。

Many AI products impress in a demo and fail in production. Drawing on Cuofano's (2026) AI Product Engineering, this study asks how the product discipline must be rebuilt when its core component is a probabilistic model that is rented and frequently replaced. A structured reading of the volume's chapters and cases yields six hypotheses drawn from its own falsifiers. Four findings emerge. First, **evaluation is the specification**: a business number, a golden set signed by a domain expert and an evaluation pipeline replace the requirements document. Second, observability, the data foundation, orchestration and governance form the instruments of production, closing the three gaps behind most failures. Third, the **absorption line** sorts the roadmap into perishable compensations and durable requirements; durable value lies in the firm's own standard, cases, data and junctions, delivered through human and agent surfaces. Fourth, intelligence has brought **marginal cost** back to software, making gross margin a design outcome and pushing pricing from seats toward usage and outcomes. Eight public cases support these mechanisms; limits and a research agenda close the study.

关键词 AI 产品工程；评估即规格；可观测性与治理；吸收线；双界面；智能边际成本

KEYWORDS AI product engineering; evaluation as specification; observability and governance; absorption line; two surfaces; marginal cost of intelligence

研究问题 *Research Questions*

RQ1	当核心组件是概率性、租用的模型时，什么取代需求文档成为产品“规格”?	With a probabilistic, rented model at the core, what replaces the requirements document as the specification?
RQ2	哪些产品工具能弥合演示与生产的缺口，并应对模型、数据与提示词的漂移?	Which instruments close the demo-to-production gap and handle drift in models, data and prompts?
RQ3	在模型持续吸收能力的条件下，产品的持久价值位于何处?	Where does durable value lie while the model keeps absorbing capability?
RQ4	智能的边际成本如何重塑毛利率、定价与在位者的商业模式?	How does the marginal cost of intelligence reshape margin, pricing and incumbents?

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第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

AI 产品的失败，较少发生在模型本身，更多发生在“能跑通的演示”与“可辩护的生产系统”之间。本章把这一断层确立为本研究的核心问题。

AI products fail less often at the model than in the gap between a demo that works and a production system that can be defended. This chapter establishes that gap as the study's central problem.

中文

ENGLISH

1.1 从演示到生产的断层

The Break Between Demo and Production

研究底本描述了一个在 2024—2026 年周期中反复出现的模式：项目从“选哪个模型”开始，功能随之堆叠，演示在受控环境中运行良好，管理层签字放行；而投产几周之后，没有人能说清 AI 在做什么、为什么这样做。底本把这类失败归结为三个缺口——**可观测性、评估与治理**——而非模型能力不足。本研究以这一诊断为起点，追问：什么样的产品学科能够系统性地弥合这些缺口？

The source volume describes a pattern that recurred throughout the 2024–2026 cycle. A project starts by choosing a model, features follow, a demo runs well in a controlled environment, and leadership signs off. A few weeks into production, nobody can say what the AI is doing or why. The volume traces these failures to three gaps, **observability, evaluation and governance**, rather than to weak models. This study takes that diagnosis as its starting point and asks what kind of product discipline can close the three gaps systematically rather than case by case.

问题的根源不在于缺少某种新工具，而在于产品的“对象”变了。产品管理是为确定性软件建立的：软件按指令行事，需求可以用文字描述，正确性交给测试。AI 时代的产品则以一个概率性的、从供应商处租用的、大约每六周就会被替换一次的模型为核心。它的行为可以在无人修改产品的情况下发生变化，而核心投入的供应商同时也可能是竞争者。

The root of the problem is not a missing tool but a changed object. Product management was built for deterministic software: the software did what it was told, requirements could be written in prose, and correctness was left to testing. The AI-era product is built around a model that is probabilistic, rented from a supplier and replaced roughly every six weeks. Its behaviour can change when nobody has changed the product, and the supplier of its core input may also be its competitor.

1.2 三重反转与一次经济断裂

Three Inversions and One Economic Break

底本将这一转变概括为三重反转；其下还有一次经济断裂：自套装软件以来，产品的核心投入第一次具有了真实的、且分布极不均衡的边际成本。

The source summarises the shift as three inversions, and beneath them an economic break: for the first time since packaged software, the product's core input carries a real, and very unevenly distributed, marginal cost.

1. **规格变成了测试套件**：无法用文字描述的概率性行为，只能以经过裁定的真实案例来界定，成功标准须在动工前以数字写明。

1. **The spec became the suite.** Probabilistic behaviour that prose cannot describe is specified by adjudicated real cases, with success stated in numbers before anything is built.

2. **产品有了两个界面**：越来越多的“用户”是代理，它们阅读价格页、调用接口、在无会话的情况下完成流程。
3. **产品在脚下漂移**：模型按供应商节奏变化，数据按业务节奏变化，提示词按团队节奏变化。

2. **The product has two surfaces.** A growing share of users are agents that read the pricing page, call the endpoint and complete workflows without a session.
3. **The product drifts underneath you.** The model changes on the vendor's schedule, the data on the business's, and the prompts on the team's.

图 1-1 产品管理的四个时代及其遗留工具
Figure 1-1 *Four eras of product management and what each left behind*



来源 Source: 据 Cuofano (2026) 第一章整理 / adapted from Chapter I.

1.3 由外而内与由内而外
Outside-In versus Inside-Out

底本用“由外而内／由内而外”解释产品工作的顺序为何必须倒转。互联网是由外而内的技术：最先改变的是分销，多年后才触及运营模式，产品经理因此从界面、漏斗与发布做起。AI 则由内而外：最先改变的是工作如何完成——什么被自动化、哪里仍由人决定；商业模式随后变化，分销最后变化。

The source uses the contrast between outside-in and inside-out to explain why the order of product work must flip. The web was outside-in: distribution changed first and the operating model was reached years later, so web-era product managers started at the interface, the funnel and the launch. AI works inside-out: the first thing to change is how work gets done, what is automated and where a person still decides; the business model follows, and distribution changes last.

核心洞见 KEY INSIGHT

从运营核心向外设计 *Engineered from the core outward*

由此推出的产品顺序是：先有测试套件，再选模型；先有章程，再有界面；先为结果定价，再做市场；人类界面与代理界面最后设计。这一顺序是本研究所有发现的组织线索。

The product order that follows is: the suite before the model, the charter before the interface, the outcome priced before it is marketed, and the human and agent surfaces designed last. This order is the organising thread of every finding in this study.

> **演示时代交付的是信心，生产时代交付的是证据。**
>
> *The demo era shipped confidence. The production era ships proof.*
>
> — Cuofano (2026), AI Product Engineering, Ch. XXXI

第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Analytical Framework

时代改变的是产品的对象，而不是产品要回答的问题。本章梳理不变的问题、新增的风险，并提出贯穿全文的分层分析框架。

The era changed the product's object, not the questions a product must answer. This chapter sets out the constant questions, the added risks and the layered framework used throughout the study.

中文　　ENGLISH

2.1 不变的问题，新增的风险

Constant Questions, Added Risks

任何产品都要回答五个问题：为谁（用户与任务）、什么价值（让任务变得更好的结果）、什么标准（“好”的定义）、如何使用（界面）、如何获利（可行性）。产品发现时代的四类风险——价值、可用性、可实现性、商业可行性——与这些问题一一对应，并在 AI 时代完整保留。

Every product answers five questions: for whom (the user and the job), what value (the outcome that makes the job better), what standard (the definition of good), how it is used (the surface) and how it pays (viability). The four risks of the discovery era, value, usability, feasibility and viability, map onto these questions and survive intact into the AI era.

底本的关键补充是两类新风险。**第五类是行为风险**：模型、数据或提示词变化后，产品能否继续做原来的事。**第六类是竞争风险**，来自产品自身的供应链：租用模型的下一个版本，会不会在不需要你的情况下完成同样的事。

The source's key addition is two new risks. **The fifth risk is behavioural**: will the product keep doing what it did when the model, the data or the prompts change? **The sixth risk is competitive** and comes from inside the product's own supply chain: will the next release of the rented model do this without you?

表 2-1 六类产品风险及其在 AI 时代的工具
Table 2-1 *Six product risks and their AI-era instruments*

风险 Risk	核心问题 Core question	AI 时代的工具 AI-era instrument
价值风险 Value	用户会想要吗? Will they want it?	以真实案例返回首个可验证结果 First verified outcome on real cases
可用性风险 Usability	用户能用起来吗? Can they use it?	人类界面 + 代理界面 Human surface plus agent surface
可实现性风险 Feasibility	我们做得出来吗? Can we build it?	以套件评测候选模型 Candidate models run against the suite
商业可行性风险 Viability	对业务成立吗? Does it work for the business?	每个被接受结果的成本 Cost per accepted outcome
行为风险（新） Behavioural (new)	变化后还能保持吗? Will it keep doing what it did?	追踪、仪表与漂移检查 Traces, gauges, drift check
竞争风险（新） Competitive (new)	下个模型会取代它吗? Will the next model do it without you?	吸收线分拣 The absorption-line sort

来源 Source: 据 Cuofano (2026) 第二章整理；工具列为本研究归纳 / adapted from Ch. II; instrument column is this study's mapping.

2.2 产品不再是模型
The Product Is Not the Model

由于核心组件是租用的、概率性的且与所有对手共享，产品不再是模型，而是企业围绕模型构建的、让结果可靠、可读、可归责的一切。底本称之为“套具”（harness）：执行工作的循环、上下文、可调用的工具与评分标准。同一模型在不同套具中处理同样工作，得分可相差两位数百分点。

Because the core component is rented, probabilistic and shared with every competitor, the product is no longer the model but everything the firm builds around it to make the outcome reliable, legible and owned. The source calls this the harness: the loop, its context, the tools it may call and the standard that grades it. The same model can score double-digit differences on the same work inside different harnesses.

概念界定 DEFINITION

评估即规格 *Evaluation as specification*

在动工、选模型、做功能之前，先以数字定义成功，并建立持续测量它的系统。黄金集即需求文档，阈值即验收标准，评估管线即测试框架；产品经理的第一件产出物是这套“裁判”。

Before any code, model or feature, define success in numbers and build the system that measures it continuously. The golden set is the PRD, the thresholds are the acceptance criteria and the pipeline is the test harness; the product manager's first artifact is this referee.

2.3 分析框架：五层结构
Analytical Framework: A Five-Layer Structure

本研究将底本重组为一个由内而外的五层框架：内层定义什么算成功，中层保证生产可见可控，外层决定谁来使用、如何获利。四个发现章依次对应其中各层。

This study reorganises the source into a five-layer framework from the inside out: the inner layers define success, the middle layer keeps production visible and controllable, and the outer layers decide who uses the product and how it pays. The four findings chapters follow these layers in turn.

图 2-1 本研究的分析框架：由内而外的五层
Figure 2-1 *The study's analytical framework: five layers from the inside out*

结果章程 Outcome charter	结果、标准、阈值、界面、边界、自主层级与负责人，一页纸	Outcome, standard, thresholds, surfaces, boundaries, autonomy tier, owner on one page
评估套件 Evaluation suite	业务数字、签署的黄金集、三层评估管线	Business number, signed golden set, three-layer pipeline
五大支柱 Five pillars	评估、可观测性、数据底座、编排、治理	Evaluation, observability, data foundation, orchestration, governance
双界面与持久性 Surfaces & durability	人类与代理两个界面；吸收线之上的持久要求	Human and agent surfaces; durable requirements above the absorption line
经济层 Economics	每个被接受结果的成本、毛利设计与定价网格	Cost per accepted outcome, designed margin, pricing grid

注 Note: 本研究据底本第 VI—XXII 章构建 / constructed by this study from Chs. VI–XXII of the source.

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究是一项基于单一底本的结构化文本分析，辅以底本所报告的、带有日期的公开案例数据。

中文

This is a structured textual analysis of a single source volume, supplemented by the dated public case data that the volume reports.

ENGLISH

3.1 研究对象与语料

Object of Study and Corpus

研究底本为 2026 年 8 月出版的第一版，分为六个部分：基础、五大支柱、技艺、团队与经济、五个动作、实践，共三十一章，配有三十七幅图版，另有四个附录——工作标准、2026 年第三季度的学科状态、一个端到端复合案例，以及八个公开案例。底本明确区分“会过期的读数”与“不会过期的工具”，这一区分也被本研究采纳。

The source is the first edition, published in August 2026. It is organised in six parts (the ground, the five pillars, the craft, the team and the economics, the five moves, and the practice), with thirty-one chapters and thirty-seven plates, plus four appendices: working standards, the state of the discipline in Q3 2026, an end-to-end composite and eight public cases. The volume explicitly separates readings that will expire from instruments that will not, and this study adopts the same separation.

31	37	8	22
章节 chapters	**图版** plates	**公开案例** public cases	**带日期的参考来源** dated references

来源 Source: Cuofano (2026), 目录与参考文献 / contents and references.

3.2 分析单位与编码程序

Units of Analysis and Coding

本研究设定三类分析单位。**机制**：底本中解释“为什么”的因果陈述，如“数据为人而建，而人会原谅错误”。**工具**：可被团队实际采用的产出物，如章程、套件、追踪、仪表页与六个计量指标。**案例**：底本附录 C 的银行复合案例与附录 D 的八个公开案例。每一个案例都对照机制与工具进行编码，记录它支持、修正或尚未检验哪一条主张。

The study uses three units of analysis. **Mechanisms** are the source's causal statements that explain why, such as "data was built for humans, and humans forgive". **Instruments** are artifacts a team can actually adopt: the charter, the suite, the trace, the gauge page and the six meters. **Cases** are the banking composite in Appendix C and the eight public cases in Appendix D. Each case is coded against the mechanisms and instruments, recording which claims it supports, qualifies or leaves untested.

方法说明 METHOD NOTE

以底本的证伪条件构造假设 *Hypotheses built from the source's own falsifiers*

底本附录 A 为每一条承重主张列出了可推翻它的观察。本研究不另造标准，而是把这些证伪条件改写为可检验的假设，从而让“方法论主张”与“经验证据”之间的距离清晰可见。

Appendix A of the source lists, for each load-bearing claim, the observation that would retire it. Rather than invent its own criteria, this study rewrites those falsifiers as testable hypotheses, which makes the distance between methodological claims and empirical evidence visible.

3.3 研究假设
Hypotheses

表 3-1 六项研究假设及其证伪条件
Table 3-1 *Six hypotheses and their falsifiers*

编号 No.	假设 Hypothesis	证伪条件 Retired if
H1 H1	先建套件的产品比先选模型的产品更高比例进入生产 Suite-first products reach production at higher rates than model-first ones	两者投产率无差异 No difference in production rates
H2 H2	有追踪的产品解决争议更快、更便宜 Traced products resolve disputes faster and more cheaply	无追踪产品同样高效 Untraced products resolve as well
H3 H3	按套件选出的模型在产品案例上优于辩论选出的模型 Models chosen against a suite beat models chosen by debate on the product's cases	辩论选择同样好 Debate does as well
H4 H4	“事故转案例”纪律减少重复事故 Incident-to-case discipline reduces repeat incidents	重复率不降 Repeat rates do not fall
H5 H5	代理型产品的消耗重尾到足以打破固定席位价 Agentic consumption is heavy-tailed enough to break flat seats	固定席位在规模上持久 Flat seats prove durable at scale
H6 H6	补偿模型缺陷的功能会在少数几次发布内被吸收 Compensations for missing capability are absorbed within a few releases	包装型功能长期存活 Wrapper features keep surviving

来源 Source: 据 Cuofano (2026) 附录 A 改写 / restated from Appendix A.

需要强调的是，底本把上述主张作为“方法”陈述，而非已经过对照检验的结论。本研究的贡献在于把它们系统化、与案例证据并置，并指出哪些假设已有初步支持、哪些仍待独立数据检验（见第八、九章）。

It should be stressed that the source states these claims as method, not as conclusions that have passed controlled tests. This study's contribution is to systematise them, set them beside the case evidence, and identify which hypotheses have early support and which still await independent data (see Chapters 8 and 9).

第 4 章 · CHAPTER 4

04 发现一：评估即规格

Finding 1: Evaluation as Specification

第一项发现回答研究问题一：取代需求文档的，是一套以数字定义成功、由领域专家签署、并在生产中持续生长的评估套件。

中文

The first finding answers RQ1: what replaces the requirements document is an evaluation suite that defines success in numbers, is signed by a domain expert and keeps growing in production.

ENGLISH

4.1 规格的三个组成部分

The Three Parts of the Specification

底本把规格拆为三部分。**数字**：不是笼统的“准确率”，而是业务真正需要的那个数——支持类产品须达到的分流率、可容忍的误报率、可承受的时延。**黄金集**：从领域专家处收集的真实案例，包括人工客服实际如何回答、灰色地带如何处理，通常六十到数百个，经裁定并签署。**管线**：把线上响应与黄金集比对、打分、低分转人工，并把纠正后的案例加回集合。

The source splits the specification into three parts. **The number** is not generic accuracy but the figure the business needs: the deflection rate a support product must reach, the false-positive rate it can tolerate, the latency it can bear. **The golden set** is real cases from domain experts, including what human agents actually answered and how grey areas were handled, usually sixty to a few hundred, adjudicated and signed. **The pipeline** compares live responses with the set, scores them, routes low scores to a person and adds corrected cases back.

套件因此是“活的系统”：每个版本冻结一次，作为循环无法左右的裁判；版本之间只通过有记录的新增扩充，绝不静默修改。行为层尤其关键：每次回答多出三次重复数据库调用，演示中无人察觉，在生产流量下却会成倍放大成本。

The suite is thus a living system: frozen per version as a referee the loop cannot shape, and grown between versions only by logged addition, never by silent edit. The behavioural layer is critical: three duplicate database calls per answer go unnoticed in a demo and multiply cost at production volume.

图 4-1 评估的三个层次：架构选择属于产品经理

Figure 4-1 *The three layers of evaluation: the architecture is the product manager's choice*

确定性层 Deterministic	格式、模式、实体识别、个人信息检测；便宜、经典，应最先处理	Formats, patterns, entity recognition, PII detection; cheap, classical, done first
语义层 Semantic	依据性、相关性、安全性；由独立的评审模型按黄金集判断并控制其偏差	Groundedness, relevance, safety; judged by a separate model anchored to the golden set
行为层 Behavioural	是否以正确次数调用正确工具、有无循环；多数团队遗漏的一层	Right tools, right number of calls, no loops; the layer most teams miss

来源 Source: Cuofano (2026), Ch. VI.

4.2 最后选择模型，并持续更换
The Model Chosen Last, and Changed Often

有了套件，模型选择从观点之争变成测量问题：演示时代以数周的“选哪家”争论开场，生产时代在八周中的第七周把候选者放到套件上跑，一天即可决定。模型更替用同一工具：新版本先以事先声明的容差在自有套件上测试，在供应商基准上更好、在产品案例上更差者，差异未解释前不予采用。同一套件在第二家供应商上运行（“接口测试”），证明方法在产品手里。

With a suite in place, model choice turns from opinion into measurement: demo-era projects opened with weeks of vendor debate, while production-era projects run candidates against the suite in week seven of eight and decide in a day. Model change uses the same instrument: a new release is tested on the product's suite against a tolerance stated in advance, and a model better on the vendor's benchmark but worse on the product's cases is not adopted until the gap is explained. Running the suite on a second provider, the joint test, shows the product holds the method.

研究发现　FINDING

基准告诉你试什么，套件告诉你留什么 *Benchmarks say what to try; the suite says what to keep*

供应商基准对产品自己的案例几乎没有信息量。只有自有套件能把“模型选择”和“模型更替”变成同一种可重复的决策，从而回应第五类（行为）风险。

Vendor benchmarks say almost nothing about a product's own cases. Only an owned suite turns model selection and model change into the same repeatable decision, which is the product's answer to the fifth, behavioural risk.

4.3 复合案例：一家零售银行的客服机器人
Composite Case: A Retail Bank's Chatbot

底本附录 C 基于公开的实践者陈述，构建了一个不具名但步骤真实的复合案例：银行客服每月约两万通来电，约六成为可分流的简单问题。首个概念验证耗资八万五千、历时六个月后失败——没有人说得清它为何出错、衡量什么、由谁负责。

Appendix C of the source builds an unnamed composite from a public practitioner account, every step real. The bank took roughly twenty thousand calls a month, about sixty percent of them simple and deflectable. A first proof of concept consumed eighty-five thousand and six months and failed: nobody could say why it was wrong, what it measured or who was accountable.

20,000	60%	200	85%
每月来电量 calls per month	**可分流的简单问题** simple, deflectable	**黄金集真实案例** golden-set cases	**准确率目标** accuracy target

来源 Source: Cuofano (2026), Appendix C（据公开实践者陈述的复合案例 / composite from a public practitioner account）.

图 4-2　重建顺序：按工作实际发生的先后
Figure 4-2 *The rebuild, in the order the work actually happens*

01	02	03	04
第 1–2 周 Weeks 1–2	**第 3–6 周** Weeks 3–6	**第 7–8 周** Weeks 7–8	**上线后第 6 周** Launch + 6 wks
200 个案例裁定成黄金集；以数字定义成功 200 cases adjudicated; success set in numbers	接通账户系统；首测即采集追踪 Account APIs wired; traces from first test	候选模型跑套件，一天内选定 Candidates run on suite; chosen in a day	利率政策变更引发漂移；修复并加案例 Policy change causes drift; fixed, case added

来源 Source: Cuofano (2026), Appendix C.

第 5 章 · CHAPTER 5

05 发现二：让生产可见、可控、可问责

Finding 2: Making Production Visible, Controllable and Accountable

第二项发现回答研究问题二：可观测性、数据底座、编排与治理共同构成生产化的仪表体系，并通过“事故转案例”的闭环应对漂移。

The second finding answers RQ2: observability, the data foundation, orchestration and governance together form the instrument set of production, and an incident-to-case loop handles drift.

中文

ENGLISH

5.1 可观测性：以一次争议为理由

Observability: Justified by a Dispute

底本用一次透支费争议说明追踪为何是产品工具而非仅是工程工具。客户认为一笔透支费收错了；产品依次完成意图分类、调取账户、检索政策文件、推理、检查护栏并作答。若每一步都被追踪——检索了什么、耗时多久、置信度如何——争议就能以“AI 看到了什么、做了什么”来回答；若没有，企业只能以折扣息事宁人。多个司法辖区的监管者已把前一种能力作为部署的前提。

The source uses an overdraft-fee dispute to show why tracing is a product instrument and not only an engineering one. A customer believes a fee was charged wrongly. The product classifies the intent, retrieves the account, pulls the policy documents, reasons, checks its guardrails and answers. If every step is traced, including what was retrieved, how long it took and with what confidence, the dispute can be answered with what the AI saw and did. If not, the firm settles with a discount. Regulators in several jurisdictions now require the first outcome as a condition of deployment.

5.2 数据底座：数据为人而建，而人会原谅

The Data Foundation: Built for Humans, Who Forgive

据实践者报告，数据底座会消耗生产项目约六成的时间。原因在于：人发现报表里的错误数字会请人更正，代理则会自信地给出错误答案，直到客户发现为止。数据质量因此从“卫生问题”变为产品的正确性本身。底座分为两半：**问题数据**——产品作答所依据的来源、系统与文档，须保持最新、受权限控制，并被描述得足以让读取目录的模型理解；**追踪数据**——轨迹、评估与反馈，需要自己的模式、存储与使用者。

Practitioners put the data foundation at about sixty percent of a production project's time. The reason fits in a sentence: a person who finds a wrong number in a report asks someone to fix it, while an agent answers confidently from it until the customer notices. Data quality thus moves from hygiene to the product's correctness. The foundation has two halves. **Question data** is everything the product reasons over, which must be current, permissioned and described richly enough for a model reading a catalog entry to understand it. **Tracking data** is the traces, evaluations and feedback, which need their own schema, storage and consumers.

~60%
数据底座占用的项目时间
project time on data

47
测试中拦截的个人数据暴露
PII exposures caught in testing

100%
追踪覆盖的应达水平
required trace coverage

来源 Source: Cuofano (2026), Chs. VIII, X, XXX（实践者报告，2026 / practitioner reports, 2026）.

案例观察　CASE OBSERVATION

陈旧底座的失败　*The stale-substrate failure*

政策已经变更并通知了客户，机器人却仍按旧文档作答，因为新文档从未进入检索库。问题之所以被发现，是因为追踪显示了读取的是哪份文档。产品的数据就是产品的行为。

The policy had changed and customers had been notified, yet the chatbot kept answering from the old document because the new one never reached the retrieval store. It was caught only because the traces showed which document had been read. The product's data is the product's behaviour.

5.3　编排：一个代理无需它，五个代理全都需要

Orchestration: One Agent Needs None, Five Need All

当多个工作单元协同时，编排成为产品决策，因为它决定了产品可能在哪里失败。底本归纳了三种覆盖多数产品的模式，并把状态管理（步骤间保留什么、存在哪里）、容错（重试、补偿、熔断）以及“决策节点图”（哪些决定可自动、哪些须上报、人在何处）列为编排内部的产品决策。

Once several working units must cooperate, orchestration becomes a product decision, because it decides where the product can fail. The source identifies three patterns that cover most products, and lists the product decisions inside orchestration as state (what persists between steps, and where), fault tolerance (retries, compensation, circuit breakers) and the junction map (which decisions may be automated, which escalate, and where the human sits).

表 5-1　三种编排模式的比较
Table 5-1　*Three orchestration patterns compared*

模式 Pattern	运作方式 How it works	优势 Strength	代价 Cost
协调者—工作者 Orchestrator-worker	一个协调者规划、分派并接收所有结果 One coordinator plans, dispatches and receives every result	集中控制，出错时只需看一处 Central control; one place to look	协调者成为瓶颈 Coordinator can bottleneck
编舞式 Choreography	独立代理订阅共享事件流并行行动 Independent agents act in parallel on a shared event stream	时延低、无中心瓶颈 Lower latency; no central bottleneck	更难推理与排错 Harder to reason about
人在回路 Human-in-the-loop	置信度低于阈值或越过决策节点时召唤人 A person is called in below a confidence threshold or at a junction	让前两者可部署 Makes the other two deployable	消耗人的注意力 Consumes human attention

来源 Source: Cuofano (2026), Ch. IX; Anthropic (2024).

5.4　治理与事故手册

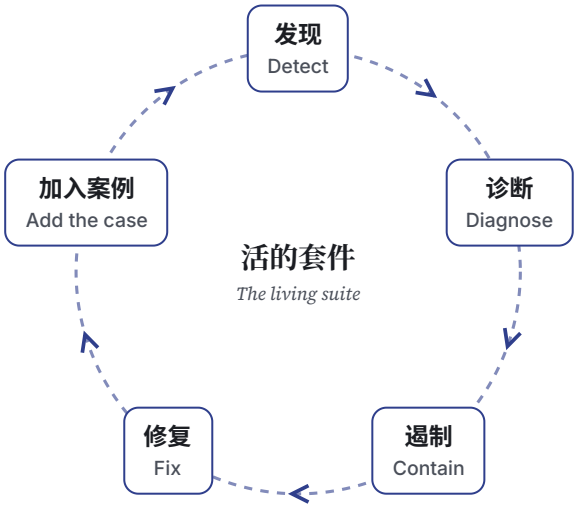
Governance and the Incident Playbook

治理回答的是“产品失败时会发生什么、由谁负责”。其组成相当具体：覆盖每一次动作与请求的审计轨迹；对每个输入运行确定性层的预校验——有团队仅在测试中就拦截了四十七次个人数据暴露；把提示词当作代码管理，每次变更记录理由；模型更替先在自有套件上测试；以及为产品行为与数据各指定一名负责人，并把事故手册接入告警系统，确保凌晨三点有人被叫醒。

Governance answers what happens when the product fails, and who answers for it. Its components are concrete: an audit trail that captures every action and request; pre-validation that runs the deterministic layer on every input, with which one team caught forty-seven personal-data exposures in testing alone; prompts managed as code, with the reason for each change recorded; model changes tested on the product's own suite; and a named owner for the product's behaviour and for its data, with an incident

playbook wired to alerting so that someone is woken at three in the morning.

图 5-1 生产事故手册：每次事故都是套件尚未拥有的案例
Figure 5-1 *The incident playbook: every incident is a case the suite did not yet have*



来源 Source: Cuofano (2026), Ch. XVIII.

手册的五步依次是：在评估仪表上发现异常；在追踪中定位到具体步骤、文档或调用；通过回滚提示词、转人工或熔断故障依赖来遏制；修复提示词、检索、工具或数据；最后把失败写入套件。评估本身的成本也须治理：每次变更只跑子集，合并时跑全量，以免裁判本身变成主要开支。

The five steps are: detect the anomaly on the evaluation dashboard; diagnose the exact step, document or call in the traces; contain by rolling back the prompt, routing to a person or tripping a circuit breaker; fix the prompt, retrieval, tool or data; and add the failure to the suite. The cost of evaluation is governed too: a subset of the suite runs on every change and the full suite on merge, so the referee does not become the main expense.

风险提示 CAUTION

七类失败，六类是三个缺口 *Seven failure families, six of them the three gaps*

底本归纳的七类产品失败中，演示型产品、先选模型之争、无规格上线、无追踪争议、陈旧底座、单界面产品这六类，本质上都是可观测性、评估或治理缺口换了一身产品的外衣；第七类是忽视吸收线。

Of the seven failure families in the source, six (the demo product, the model-first debate, the spec-less launch, the trace-less dispute, the stale substrate and the single-surface product) are the observability, evaluation or governance gap wearing a product's clothes. The seventh is the absorption line ignored.

第 6 章 · CHAPTER 6

06 发现三：吸收线、残余与双界面

Finding 3: The Absorption Line, Residue and Two Surfaces

第三项发现回答研究问题三：在模型持续吸收能力的条件下，持久价值位于企业自有的事实之中，并通过双界面抵达人与代理。

中文

The third finding answers RQ3: while the model keeps absorbing capability, durable value lies in facts the firm owns, delivered to people and agents through two surfaces.

ENGLISH

6.1 模型既是组件，也是竞争者

The Model as Component and Competitor

第六类风险需要自己的工具，因为模型不只是产品的组件，也是按计划发动进攻的竞争者。每一次供应商发布都会吸收上一季度还属于某些产品的能力。底本借用工程卷中的“移动边界”来描述这一机制：模型训练、吸收套具所补偿的能力，补偿随之被舍弃。第一幕“包装层”产品的功能——摘要、翻译、与文档对话——都曾是产品，直到模型原生具备这些能力。

The sixth risk needs an instrument of its own, because the model is not only the product's component but also its competitor, and it attacks on a schedule. Each vendor release absorbs capability that was somebody's product the quarter before. The source borrows the engineering volumes' "moving boundary" for the mechanism: the model trains, absorbs what harnesses were compensating for, and the compensations are shed. The wrapper features of the first act, summaries, translations, chatting with a document, were each a product until the model did them natively.

因此，路线图要逐项分拣成两堆。**易逝的补偿**：因模型暂时做不到而构建的脚手架、变通与能力补丁。它们值得做，但注定被删除；纪律是做得便宜、标记为易逝，并在模型吸收它们时把删除计为进展。**持久的要求**：企业的标准、经过裁定的案例、数据底座、决策节点，以及“决定了什么、为什么”的记录。它们不是能力，而是关于企业及其客户的事实，没有哪个模型版本能吸收它不拥有的事实。

The roadmap is therefore sorted, item by item, into two piles. **Perishable compensations** are the scaffolding, workarounds and capability patches built because the model cannot yet do something. They are legitimate to build and destined to be deleted; the discipline is to build them cheaply, mark them as perishable and count their deletion as progress. **Durable requirements** are the firm's standard, its adjudicated cases, its data foundation, its junctions and its record of what was decided and why. These are not capabilities but facts about the firm and its customers, and no release absorbs a fact it does not have.

图 6-1 复利发生在哪里：吸收线之上的三层

Figure 6-1 *Where compounding lives: the layers above the absorption line*

元套具 Meta-harness	跨多个循环路由工作、持有套件与共享记录、执行企业政策；由企业的判断构成	Routes work across loops, holds suites and the shared record, enforces policy; made of the firm's judgment
套具 Harness	执行循环、上下文、工具与评分标准；产品真正所在之处	Loop, context, tools and grading standard; where the product actually lives
租用模型 Rented model	可替换的引擎；按供应商节奏进步并吸收补偿	A replaceable engine; advances on the vendor's schedule and absorbs compensations

来源 Source: Cuofano (2026), Chs. V, XVI.

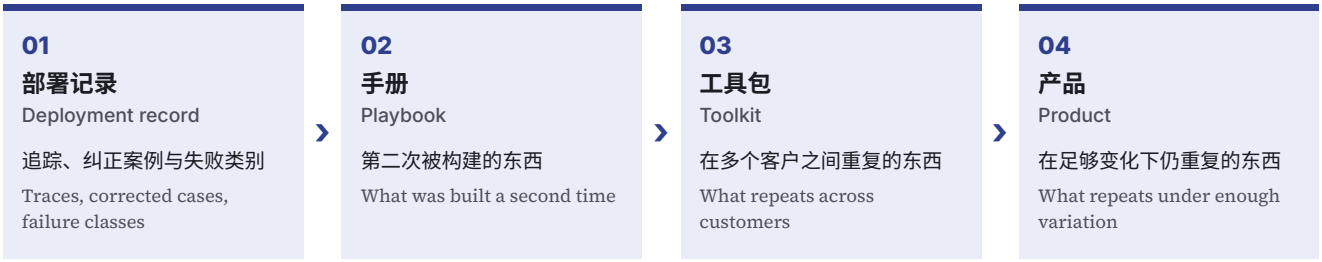
> **模型会推进到一切领域，唯独推进不到关于你企业的事实。就在那里构建。**
>
> *The model advances into everything except the facts about your firm. Build there.*
>
> — Cuofano (2026), AI Product Engineering, Ch. XVI

6.2 路线图来自残余
The Roadmap from Residue

路线图不再来自需求队列，而来自记录。每一次部署、每一条追踪、每一个被纠正的案例，都是产品应当成为什么的证据。产品经理的发现工作随之从“访谈想象中的需求”转向“部署记录中已证明的需求”，从功能请求转向套件反复捕获的失败类别。每个检查点的残余复盘只问一个问题：什么东西被构建了两次?

The roadmap stops arriving from the request queue and starts arriving from the record. Every deployment, trace and corrected case is evidence of what the product should become. Discovery work moves from interviews about imagined needs to the deployment record of demonstrated ones, and from feature requests to the failure classes the suite keeps catching. The residue review at each checkpoint asks one question: what was built twice?

图 6-2 残余的收获层级：从记录到产品
Figure 6-2 *The residue harvest hierarchy: from record to product*



来源 Source: Cuofano (2026), Chs. IV, XV.

6.3 双界面与首个可验证结果
Two Surfaces and the First Verified Outcome

产品要被设计两次：一次为看屏幕的人，一次为从不看屏幕的代理。人类界面保留上一时代的全部纪律，并把验证时刻设计为激活——激活即用户在自己真实工作上得到的第一个可验证结果。代理界面本质上是一份规格：可调用的接口、结构化报价、机器可读的条款、无需会话即可完成的动作。两个界面须抵达同一结果、由同一套件评分，代理用户群单独度量。应用层的三幕——包装层、套具、分销回归——也说明：套具是产品，模型是租户，渠道决定平局。

The product is designed twice: once for the person who sees the screen and once for the agent that never does. The human surface keeps the previous era's disciplines and designs the moment of verification as activation, defined as the first verified outcome on the user's own work. The agent surface is essentially a specification: callable endpoints, structured offers, machine-readable terms and actions completable without a session. Both surfaces must reach the same outcome, graded by one suite, with the agent cohort measured separately. The application layer's three acts, wrapper, harness and the return of distribution, make the same point: the harness is the product, the model is the tenant, and the channel decides the tie.

第 7 章 · CHAPTER 7

07 发现四：智能的边际成本与定价重构

Finding 4: The Marginal Cost of Intelligence and the Repricing

第四项发现回答研究问题四：智能让软件重新拥有了边际成本，毛利率因此成为设计结果，定价被迫从席位转向用量与结果。

The fourth finding answers RQ4: intelligence has given software a marginal cost again, so gross margin becomes a design outcome and pricing is pushed from seats toward usage and outcomes.

中文

ENGLISH

7.1 边际成本回到软件业

Marginal Cost Returns to Software

软件经济学建立在近乎为零的边际成本之上：约 80% 的毛利率是品类基线，它为销售投入提供资金、让免费层免费，也让席位价可以完全脱离成本。底本的判断是：软件卖的是拷贝，这个时代卖的是工作，而工作有销货成本——每个结果都消耗按词元计量、由供应商计费的推理。

Software economics rested on a marginal cost close to zero. Gross margins of about eighty percent were the category's baseline; they funded sales, made the free tier free and let the seat price float free of cost. The source's judgment is that software sold copies while this era sells work, and work has a cost of goods: every outcome consumes inference metered in tokens and billed by a supplier.

7.2 边际成本的三个性质

Three Properties of the New Marginal Cost

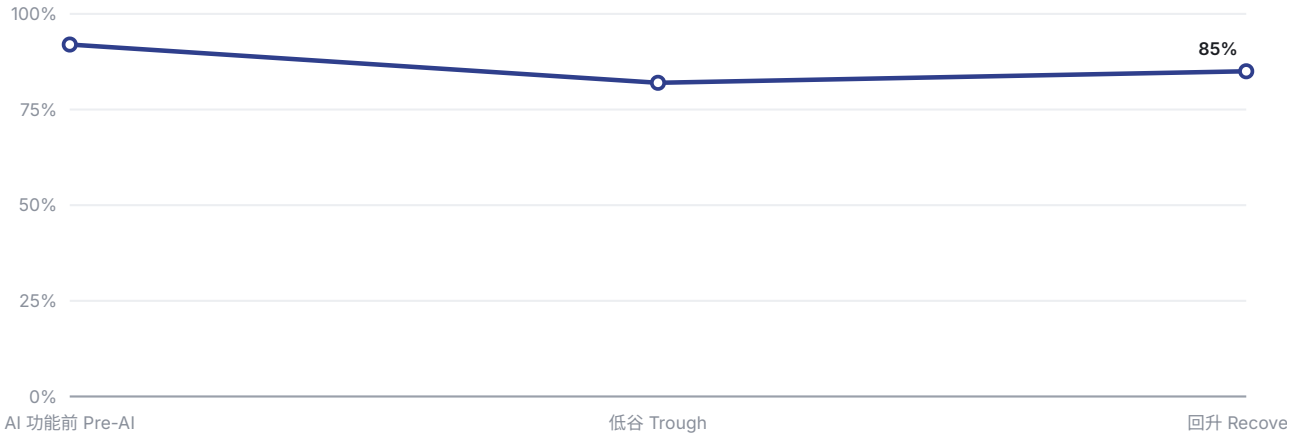
其一，同时通缩与上升。能力不变时，推理价格约每年下降十倍；但产品交付的是前沿能力，推理模型与代理循环使每个任务的词元数增长得比单价下降更快。“每词元价格崩塌、每用户账单上升”同时为真，唯一持久的预算形式是“每个结果的词元数”。

First, it deflates and rises at once. At fixed capability the price of inference falls roughly tenfold a year, but products ship the frontier, and reasoning models and agent loops multiply tokens per task faster than prices fall. The price per token collapses while the bill per user rises, and the only durable budget is tokens per outcome.

其二，毛利率成为设计结果。记录最完整的产品在租用前沿模型时都曾负毛利，随后通过路由、缓存与基于自身残余训练的自有模型恢复毛利。

Second, gross margin becomes a design outcome. The best-documented products ran negative margins while renting frontier models and recovered through routing, caching and owned models trained on their own residue.

图 7-1 Figma 毛利率：软件获得了销货成本
Figure 7-1 *Figma's gross margin: software acquires a cost of goods*

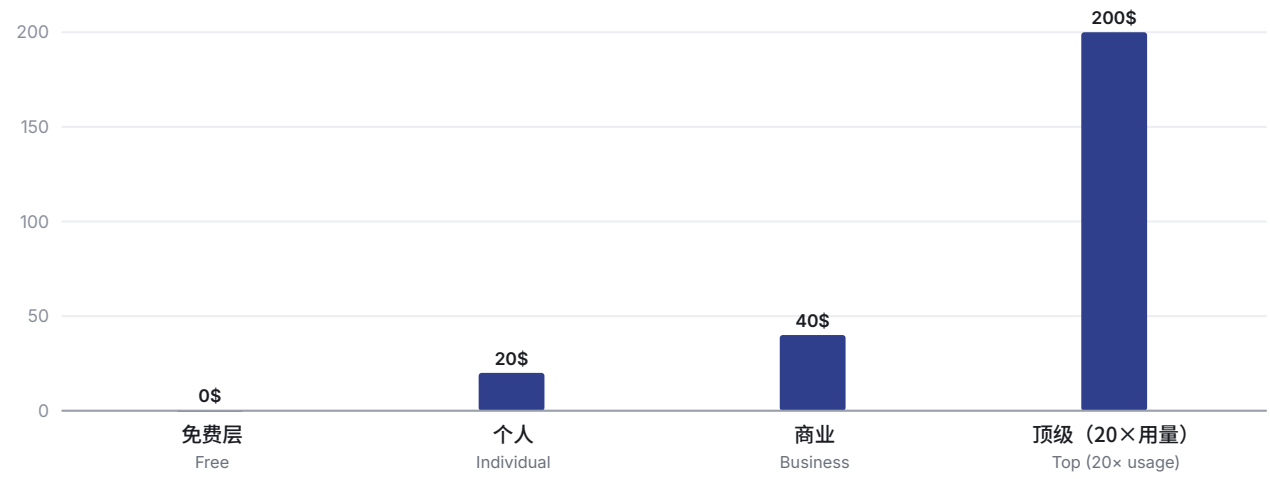


来源 Source: Cuofano (2026), Appendix D, citing Figma Q2 2026 results (August 2026). 同期收入增速由 38% 升至 48%，经营利润率收窄 / revenue growth rose from 38% to 48% while operating margin compressed.

其三，消耗是重尾的。席位价以中位用户为前提，代理型产品却没有值得据以定价的中位用户：为关键工作连续数小时运行代理的用户，消耗高出几个数量级，供应商顶级套餐定价为标准套餐的十到二十倍。席位从两端断裂：轻度用户被高估而流失，重度用户在任何固定价格下都无利可图。

Third, consumption is heavy-tailed. The seat price assumed the median user, but agentic products have no median user worth pricing to: users running agents for hours on critical work consume orders of magnitude more, and vendors price top plans at ten to twenty times the standard one. The seat breaks from both ends: the light user is overpriced and churns, the heavy user is unprofitable at any flat price.

图 7-2 Cursor 的套餐分层：把重尾消耗定价
Figure 7-2 *Cursor's tiers: pricing the heavy tail*



来源 Source: Cuofano (2026), Appendix D, citing Sacra (2026) third-party estimates.

7.3 定价跟随工具
Pricing Follows the Instrument

定价如今是产品决策，因为被定价的对象是被设计出来的。底本的网格是：辅助人时按席位，归因不清地自主行动时按用量，拥有可验证结果时按结果；能采用哪一档取决于产品能验证什么。本研究将其整理为图 10 的矩阵。

Pricing is now a product decision, because the object priced is designed. The source's grid is seats where the product assists, usage where it acts without clean attribution, and outcomes where it owns a verifiable result; which tier is possible depends on what the

product can verify. Figure 10 arranges the grid as a matrix.

图 7-3 定价网格：自主程度 × 结果可验证性（本研究整理，示意）
Figure 7-3 *The pricing grid: autonomy × verifiability (arranged by this study, illustrative)*



注 Note: 三档来自 Cuofano (2026) Ch. XIV；矩阵排列与"席位+额度"象限据附录 D 的 Figma 案例归纳 / tiers from Ch. XIV; arrangement and the hybrid quadrant drawn from the Figma case.

7.4 在位者的重新定价
The Incumbent's Repricing

行业大部分收入仍在按席位定价的在位者手中。席位卖访问，代理卖工作，而工作可以计量。底本指出三重碰撞。**计量**：最大的 CRM 厂商不到两年内为代理产品换了三种计价方式，——每次对话 2 美元、每 10 万额度 500 美元的 Flex Credits、每月 125 美元起的按用户许可——且三者同时有效。**算术**：代理调用逐季倍增而人工使用持平，中间件与仪表板收入线最先转负。**界面**：在位者须在守一扇越来越少人打开的前门与成为每个代理都要经过的受治理底座之间选择。

Most of the industry's revenue still sits in incumbents priced per seat. The seat sold access; the agent sells work, and work can be metered. The source names three collisions. **The meter**: the largest CRM vendor cycled through three pricing models for its agent product in under two years, (per conversation, Flex Credits at 00 per 100,000, per-user licences from 25 a month), all live at once. **The arithmetic**: agent calls multiplied quarter over quarter while human usage stayed flat, and middleware and dashboard lines turned negative first. **The interface**: the incumbent must choose between a front door fewer people open and becoming the governed substrate every agent passes through.

研究发现 FINDING

席位卖访问，代理卖工作 *The seat sold access; the agent sells work*

以访问定价的存量用户基础，无法在不重新为自己定价的情况下为工作定价。GitHub Copilot 在同一时期从固定席位转向计量请求，再转向以词元计的额度，说明这并非巨头独有的现象。成本落在租用的核心上，利润只能来自企业自己添加的东西。

An installed base priced on access cannot price work without repricing itself. GitHub Copilot moved in the same period from a flat seat to metered requests and then to token-denominated credits, showing this is not confined to giants. The cost lives in the rented core; the margin can only come from what the firm adds.

第 8 章 · CHAPTER 8

08 案例比较分析：八个公开案例

Comparative Case Analysis: Eight Public Cases

本章把底本附录 D 的八个公开案例作为产品决策重新编码，检验前四项发现在不同层级上是否成立。

This chapter recodes the eight public cases in Appendix D of the source as product decisions and checks whether the four findings hold at different layers of the stack.

中文

八个案例横跨底本的整张地图：应用层、模型层自有产品、消费者问答引擎、垂直领域、平台轨道、在位者与企业决策节点。底本强调，数字来自公开报道与公司披露并注明日期，年化收入为报道时的快照，第三方估算另有标注；“解读”属于底本，本研究在此基础上做二次编码。

ENGLISH

The eight cases span the source's whole map: the application layer, the model layer's own products, the consumer answer engine, the vertical, the platform rail, the incumbent and the enterprise junction. The source stresses that its numbers come from dated public reporting and company disclosures, that run-rates are annualised snapshots as reported, and that third-party estimates are flagged. The reading is the source's; this study adds a second round of coding on top of it.

表 8-1 八个案例：关键产品决策与对应发现
Table 8-1 *Eight cases: the decisive product move and the finding it bears on*

案例 Case	关键产品决策 Decisive product move	对应发现 Finding
Cursor Cursor	以自身用量训练自有模型并路由，扭转负毛利 Owned model on own usage plus routing reversed negative margin	发现一、四 F1, F4
Claude Code Claude Code	无图形界面，产品完全是第二界面 No GUI: the product is entirely the second surface	发现三 F3
Codex Codex	开放套具，模型可替换 Open harness; model not hardcoded	发现一、三 F1, F3
Perplexity Perplexity	放弃广告，订阅加用量计费 Dropped ads; subscription plus usage layer	发现四 F4
Harvey Harvey	退役自有微调模型，保留垂直纹理 Retired its fine-tuned model; kept the vertical grain	发现三 F3
Shopify Shopify	结构化目录与结算轨道 Structured catalogue and settlement rail	发现三 F3
Figma Figma	在席位上叠加 AI 额度 AI credits bolted onto seats	发现四 F4
Palantir Palantir	按结果定价，持续换测模型 Outcome pricing; continuous model swap-testing	发现一、四 F1, F4

来源 Source: Cuofano (2026), Appendix D and Notes; coding by this study.

8.1 利润率作为产品决策
Margin as a Product Decision

Cursor 的公开记录是“残余循环”与“模型接口”决定产品经济性的最清晰案例。它在相当长时期内处于负毛利，2025 年 11 月推出基于自身使用数据训练的模型，并把便宜的工作路由给便宜的模型；据第三方估算，到 2026 年 4 月实现了微弱的毛利盈利，企业账户为正、个人账户仍亏损。2026 年 2 月推出的云端代理产品被报道认为推动了年化收入从 2 月的 20 亿美元增至 6 月的 40 亿美元。Palantir 则处在定价网格的顶端：按已安装的结果定价，收入同比接近翻倍、净美元留存率超过 150%，资本开支不足收入的 1%，并以专门工具持续对模型做替换测试。

Cursor's public record is the clearest case of the residue loop and the model joint deciding a product's economics. It ran negative gross margins for much of its life, shipped a model trained on its own usage in November 2025 and began routing cheaper work to cheaper models. By April 2026, per third-party estimates, it had reached slight gross-margin profitability, with enterprise accounts positive and individual accounts still loss-making. A cloud-agent launch in February 2026 was credited with part of the rise in run-rate from two billion dollars in February to four billion by June. Palantir sits at the top of the grid: it prices installed outcomes, with revenue nearly doubling, net dollar retention above 150 percent, capex under one percent of revenue, and continuous swap-testing of models.

8.2 第二界面即产品
The Second Surface as the Product

Claude Code 没有任何图形界面，完全是从命令行与开发者工具中触达的第二界面。据公司披露，它在上线六个月内达到 10 亿美元年化收入，到 2026 年 2 月超过 25 亿美元；一项约四十万次会话的使用研究显示用户每周使用约二十小时。底本提醒，这二十小时应读作“结果产出率”而非“参与度”。Codex 则把套具分层开放：循环开源、默认模型不硬编码、指令约定可被其他代理采用。2026 年 2 月桌面应用发布后，周活跃用户增至 6 月的五百万以上，知识工作者约占五分之一且增长快三倍。Shopify 把商户目录结构化、让代理意图可以完成结算：AI 来源流量同比增至三倍，结构化目录数据的转化率是抓取页面的两倍，四分之三的代理订单落在前一百个品类之外。

Claude Code shipped with no graphical interface at all; it is entirely the second surface, reached from the command line and developer tools. Per company disclosure it reached one billion dollars in annualised revenue within six months and over 2.5 billion by February 2026, and a usage study of about 400,000 sessions reported roughly twenty hours a week in the tool, which the source reads as outcome rate rather than engagement. Codex opened its harness in layers: an open-source loop, models not hardcoded, an instruction convention others could adopt. After its desktop app shipped in February 2026, weekly active users passed five million by June, with knowledge workers about a fifth of users and growing three times faster. Shopify structured merchant catalogues and wired checkout so agent intent can clear: AI-originated traffic tripled, structured catalogue data converted at twice the rate of scraped pages, and three quarters of agent orders fell outside the top hundred categories.

$2.5B+
Claude Code 年化收入（2026.2）
Claude Code run-rate, Feb 2026

5M+
Codex 周活跃用户（2026.6）
Codex weekly actives, Jun 2026

3×
Shopify AI 来源流量同比
Shopify AI traffic, YoY

>150%
Palantir 净美元留存率
Palantir net dollar retention

来源 Source: Cuofano (2026), Appendix D and References 5, 8, 20, 22; annualised snapshots as reported, Q3 2026 reading.

8.3 吸收与重新定价
Absorption and Repricing

Harvey 是吸收线穿过一个在售产品的公开实例：它早期构建了法律微调模型，在前沿推理模型直接吸收法律推理后将其退役。留存下来的是底本所说的全部持久要求——在美国百强律所中约半数的采用、保险与金融服务等行业模块、与文档管理系统的集成。年化收入在“删除”之后继续复利：2025 年 8 月为 1 亿美元，2026 年 1 月约翻倍，2026 年年中据第三方估算约为三倍。Perplexity 在 2026 年 2 月完全放弃广告、推出自主代理并叠加用量计费，据报道 3 月年化经常性收入超过 4.5 亿美元；Figma 则在毛利率从 92% 回落后，以 AI 额度和第一方代理作为成本杠杆。

Harvey is the absorption line crossing a shipping product in public. It built a fine-tuned legal model early and retired it when frontier reasoning models absorbed legal reasoning outright. What survived was everything the source calls durable: adoption across about half of the largest hundred American law firms, sector modules for insurance and financial services, and integrations into document-management systems. Revenue kept compounding through the deletion: 100 million dollars annualised in August 2025, roughly double by January 2026 and about triple by mid-2026 per third-party estimates. Perplexity dropped advertising in February 2026, launched an autonomous agent and added usage pricing, with ARR reported above 450 million dollars in March; Figma, after its margin fell from 92 percent, used AI credits and a first-party agent as cost levers.

案例观察 CASE OBSERVATION

八个案例的共性 *What the eight share*

模型在接口之后被替换、混用或删除；残余变成了自有模型、开放架构或受治理的记录；第二界面作为产品而非功能上线；路线图按吸收线分拣。没有一个案例是“做完演示然后等待”。这为 H5、H6 提供了初步支持，但 H1—H4 仍缺乏可比对照数据。

In each case the model was swapped, blended or deleted behind a joint; the residue became an owned model, an open architecture or a governed record; the second surface shipped as a product, not a feature; and the roadmap was sorted at the absorption line. None shipped a demo and waited. This offers early support for H5 and H6, while H1–H4 still lack comparable control data.

第 9 章 · CHAPTER 9

09 讨论、结论与未来研究方向

Discussion, Conclusions and Future Research

本章讨论四项发现对理论与实践的启示，说明研究的局限与风险，回答研究问题，并提出后续研究议程。

中文

This chapter discusses what the four findings imply for theory and practice, states the study's limits and risks, answers the research questions and proposes a research agenda.

ENGLISH

9.1 理论启示

Implications for Theory

四项发现指向同一结论：产品学科的重心从描述行为转向验证结果。由此带来三点理论修正：规格从文字变为经裁定的案例，产品经理的深层能力转向构建裁判；产品边界从模型移到套具与元套具，优势来源从能力转向企业自有的事实；经济性与吸收线是同一课题的两面——成本在租用的核心，利润在一切自有之物。

The four findings point to one conclusion: the product discipline's centre of gravity has moved from describing behaviour to verifying outcomes. Three theoretical revisions follow: the specification changes from prose to adjudicated cases, and the product manager's deep skill moves to building the referee; the product boundary moves from the model to the harness and meta-harness, and advantage moves from capability to facts the firm owns; and the economics and the absorption line are one lesson, with cost in the rented core and margin in everything owned.

9.2 实践启示：五个动作与六个计量指标

Implications for Practice: Five Moves and Six Meters

底本把学科压缩为五个动作：**为结果立章程**、**冻结规格**、**为每个决策布设仪表**、**交付双界面**、**管理变更**。团队围绕套件组建，领域专家是签署黄金集的正式成员，评估负责人是专职角色。判断成效的六个指标应一起按趋势读取：

- 套件通过率（相对阈值，按版本）
- 魔法时间（激活时钟）
- 追踪覆盖率（低于 100% 即有未设防区域）
- 事故转案例率（一周内进入套件的比例）
- 漂移容差（最近一次换模型的套件差值）
- 每个被接受结果的成本（按档位读取）

The source compresses the discipline into five moves: **charter the outcome**, **freeze the spec**, **instrument every decision**, **ship the two surfaces** and **manage the change**. The team forms around the suite, with the domain expert as a full member who signs the golden set and a named evaluation owner. Six meters, read together and on trend, judge whether the moves work:

- Suite pass rate against threshold, per version
- Time-to-magic, the activation clock
- Trace coverage; below 100% leaves undefended regions
- Incident-to-case rate within the week
- Drift tolerance across the last model change
- Cost per accepted outcome, read per tier

> **为结果立章程。冻结规格。交付双界面。**
>
> *Charter the outcome. Freeze the spec. Ship the two surfaces.*
>
> — Cuofano (2026), AI Product Engineering, Epilogue

9.3 局限与风险
Limitations and Risks

风险提示 CAUTION

解读本研究时需注意的四点 *Four cautions in reading this study*

（1）单一底本：框架与案例解读出自同一作者体系，有确认偏差之虞。（2）证据层级：部分数字为第三方估算或报道快照，案例偏向成功者。（3）时效：读数截至 2026 年第三季度。（4）章程形式、双界面规则、吸收分拣与六个指标属方法陈述，尚未经对照检验。

(1) Single source: framework and case readings come from one author's system, risking confirmation bias. (2) Evidence tiers: some figures are third-party estimates or reported snapshots, and the cases skew toward successes. (3) Timeliness: readings are as of Q3 2026. (4) The charter form, two-surfaces rule, absorption sort and six meters are stated as method, untested against controls.

9.4 结论
Conclusions

对四个研究问题的回答是：（RQ1）评估套件取代需求文档；（RQ2）四类支柱构成生产仪表，以“事故转案例”闭环应对漂移；（RQ3）持久价值位于吸收线之上，即企业的标准、案例、数据、决策节点与记录，并经由双界面交付；（RQ4）智能的边际成本使毛利率成为设计结果，推动定价沿“席位—用量—结果”迁移，并迫使在位者重新定价。

The answers to the four questions are: (RQ1) an evaluation suite replaces the requirements document; (RQ2) four pillars form the instruments of production, with an incident-to-case loop handling drift; (RQ3) durable value sits above the absorption line, in the firm's standard, cases, data, junctions and record, delivered through two surfaces; (RQ4) the marginal cost of intelligence makes gross margin a design outcome, moves pricing along the seat–usage–outcome grid and forces incumbents to reprice.

9.5 未来研究方向
Future Research Agenda

1. 以多企业样本检验 H1：比较“先套件”与“先模型”项目的投产率。
2. 量化追踪对争议解决成本的影响（H2），尤其在受监管行业。
3. 建立吸收线的纵向数据集：追踪补偿型功能从上线到被吸收的发布次数（H6）。
4. 测量代理型产品的消耗分布与各档位毛利（H5），检验固定席位是否可持续。
5. 建立代理界面的度量方法：代理用户群的激活、留存与归因。

1. Test H1 on a multi-firm sample, comparing production rates of suite-first and model-first projects.
2. Quantify the effect of tracing on dispute-resolution cost (H2), especially in regulated industries.
3. Build a longitudinal dataset of the absorption line, counting releases from launch to absorption for compensation features (H6).
4. Measure consumption distributions and per-tier margins in agentic products (H5) to test whether flat seats can last.
5. Develop measures for the agent surface: activation, retention and attribution of the agent cohort.

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对研究底本《AI Product Engineering》（Cuofano，2026 年 8 月第一版）的分析性解读，而非翻译或摘要。文中的概念、机制与案例均来自底本，分析框架、假设表述与比较结构为本研究所建。所有数字均为底本所报告的、截至 2026 年第三季度的时点读数（年化收入等为报道时的快照，第三方估算已注明），预计会被后续数据取代；凡属示意性的图表均在图注中标明“示意”。

This study is an analytical reading of the source volume, AI Product Engineering (Cuofano, first edition, August 2026), not a translation or summary. Concepts, mechanisms and cases come from the source; the analytical framework, the wording of the hypotheses and the comparative structure are this study's own. All figures are the source's dated readings as of Q3 2026 (run-rates are annualised snapshots as reported, and third-party estimates are flagged) and are expected to be superseded; any chart that is schematic is marked illustrative in its caption.

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